

Displacement versus HPWL

Legalization cost combines HPWL and displacement

The idea

- Lambda one gives cost forty-two
- Higher lambda favors staying near global placement even when wirelength could be slightly

Pseudocode

- This module's pseudocode is a cost function, not a packer
- After you have a legal layout
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Plug in the Abacus goldens: thirty-eight plus lambda times four
- Lambda one costs forty-two; lambda five costs fifty-eight
- Higher lambda means the sketch favors staying near the global place

Algorithm sketch — try these

INPUT: legal positions, origin, nets, $\lambda \geq 0$

OUTPUT: cost, HPWL, disp

disp $\leftarrow \sum |\Delta x| + |\Delta y|$ vs origin

HPWL $\leftarrow \sum \text{net bbox (cell centers)}$

cost $\leftarrow \text{HPWL} + \lambda \cdot \text{disp}$

GOLDEN Abacus: HPWL=38 disp=4

$\lambda=1 \rightarrow 42$; $\lambda=5 \rightarrow 58$

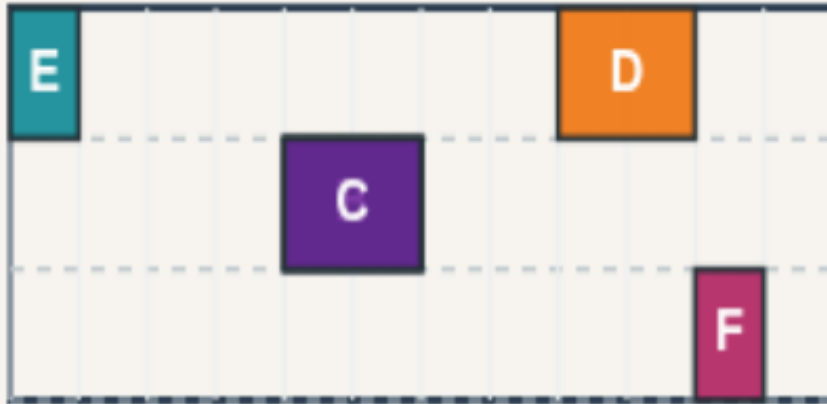
Cost = HPWL + $\lambda \cdot \text{displacement}$

DISPLACEMENT VS HPWL

Cost = HPWL + $\lambda \cdot \text{displacement}$

Step 1 / 5 · cost-formula · module03-03-displacement-hpwl

chip 12x6 · 3 rows · site 1



Explanation

Legalization objectives often combine wirelength with a displacement penalty from the pre-legalize layout. Lambda controls how hard you resist moving cells away from global placement.

- $\text{cost} = \text{HPWL} + \lambda \cdot \text{disp}$
- $\text{disp} = \text{L1 from origin}$
- Same nets as placement labs

λ tunable
report both terms

Cost = HPWL + $\lambda \cdot \text{displacement}$

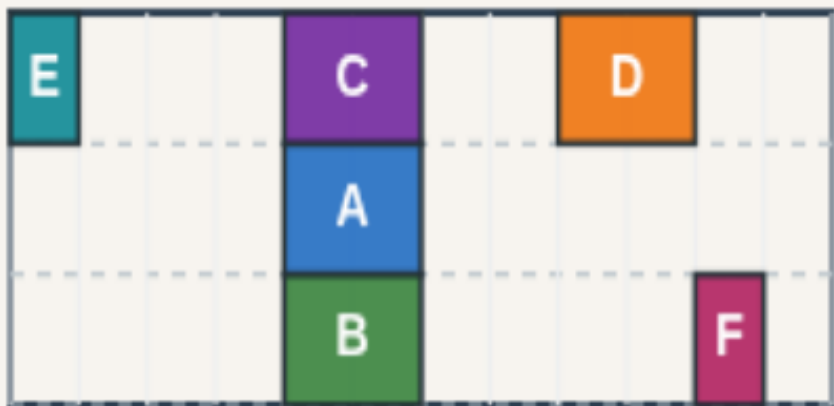
Abacus vs overlap origin

DISPLACEMENT VS HPWL

Abacus vs overlap origin

Step 2 / 5 · abacus-metrics · module03-03-displacement-hpwl

chip 12x6 · 3 rows · site 1



Explanation

Abacus on the overlap seed yields HPWL thirty-eight and displacement four versus the illegal starter. That is the Pareto point with lower movement.

- origin: OVERLAP
- abacus legal
- HPWL 38 · disp 4

HPWL: 38
disp: 4
HPWL overlap illegal: n/a

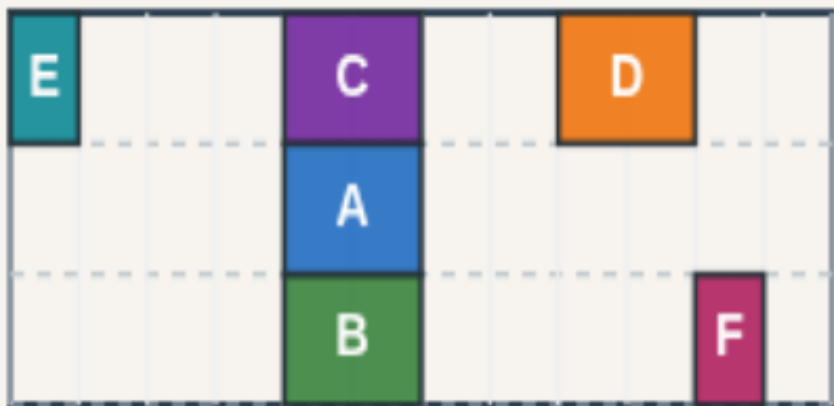
$\lambda = 1 \rightarrow \text{cost } 42$

DISPLACEMENT VS HPWL

$\lambda = 1 \rightarrow \text{cost } 42$

Step 3 / 5 · lambda-1 · module03-03-displacement-hpwl

chip 12×6 · 3 rows · site 1



Explanation

With lambda one, cost equals thirty-eight plus four equals forty-two. Displacement is cheap—one HPWL point buys one unit of movement in the objective.

- $38 + 1 \times 4 = 42$
- abacusCostLambda1 golden
- Compare to Tetris cost

$\lambda=1$ cost: 42
measured: 42

$\lambda = 1 \rightarrow \text{cost } 42$

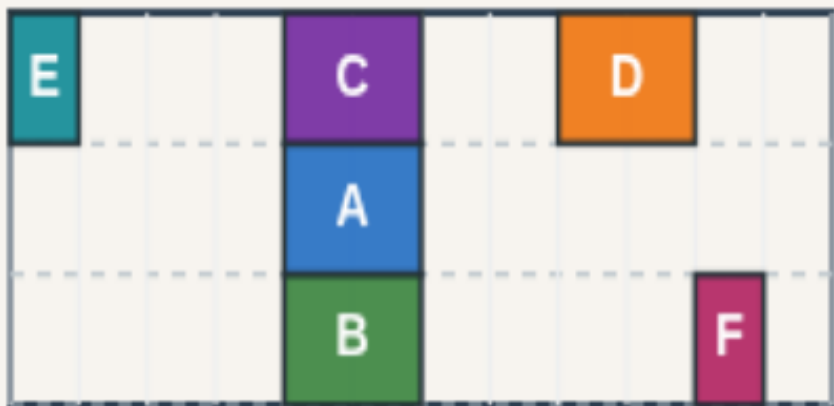
$\lambda = 5 \rightarrow \text{cost } 58$

DISPLACEMENT VS HPWL

$\lambda = 5 \rightarrow \text{cost } 58$

Step 4 / 5 · lambda-5 · module03-03-displacement-hpwl

chip 12×6 · 3 rows · site 1



Explanation

Raise lambda to five: cost becomes thirty-eight plus five times four equals fifty-eight. The same legal layout looks expensive when you punish displacement heavily.

- $38 + 5 \times 4 = 58$
- abacusCostLambda5 golden
- Higher λ favors staying put

$\lambda=5$ cost: 58
measured: 58

$\lambda = 5 \rightarrow \text{cost } 58$

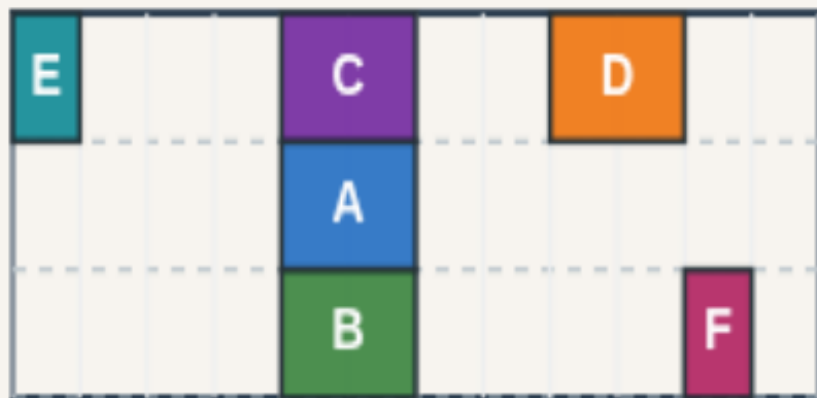
Higher λ favors staying put

DISPLACEMENT VS HPWL

Higher λ favors staying put

Step 5 / 5 · lambda-tradeoff · module03-03-displacement-hpwl

chip 12×6 · 3 rows · site 1



Explanation

When lambda is large, algorithms that minimize displacement—Abacus over Tetris—win the combined cost even if HPWL is slightly worse. Quote λ whenever you compare legalizers.

- Low λ : wirelength dominates
- High λ : displacement dominates
- Tetris: lower HPWL, higher disp

abacus: HPWL 38 disp 4
tetris: HPWL 32 disp 6
 $\lambda=5$ gap widens

Higher λ favors staying put

Browser lab track

- In the browser lab track, open the **displacement-hpwl** lab from the tools shelf
- Open the interactive lab
- Reveal golden is study-only
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny\_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module